

POLITECNICO DI TORINO

Master Degree in Sustainable nuclear energy



**Politecnico
di Torino**

Fuel cycle, waste and decommissioning

**Dynamics modeling and preliminary design of
a D-T fusion fuel cycle**

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1 Introduction

A closed tritium fuel cycle represents a fundamental requirement for the operation of a deuterium-tritium (D-T) fusion power plant (FPP). The optimization of tritium usage throughout the plant is crucial to achieve tritium self-sufficiency and to minimize tritium inventories, which has significant implications for both the economic viability and safety of the facility. This report details the preliminary design of key components within a fusion fuel cycle and the development of a dynamic model to simulate the system's behavior over time. The specific FPP under consideration is based on a compact tokamak design operating with a 50-50 D-T mixture and producing a thermal fusion power of $P_{\text{fus}} = 500 \text{ MW}_{th}$.

This work can be subdivided in three main objectives. First, to size the primary vacuum pumping system, ensuring it meets the operational demands and respect safety constraints. Second, to develop a fuel cycle model using a lumped parameter approach based on component residence times. Third, to utilize this model to simulate the plant's startup transient and steady-state operation to determine the minimum required startup tritium inventory (I_{startup}) and the required Tritium Breeding Ratio (TBR_r) necessary to meet a target doubling time of two years. The successful determination of these parameters is essential for assessing the plant's capability to be tritium self-sufficient.

2 Problem description

The fuel cycle for the compact tokamak follows a simplified layout as shown in Figure 1. The cycle integrates several major subsystems: the plasma chamber, the breeding blanket (BB), the tritium extraction system (TES), the heat exchanger (HX), the primary vacuum pumping system (cryopumps, CP), the tritium processing plant (P), and the storage and delivery system (SDS). The dynamic evolution of tritium inventories within these components is modeled using a lumped parameter approach, where the key parameter for each component is its characteristic residence time (τ).

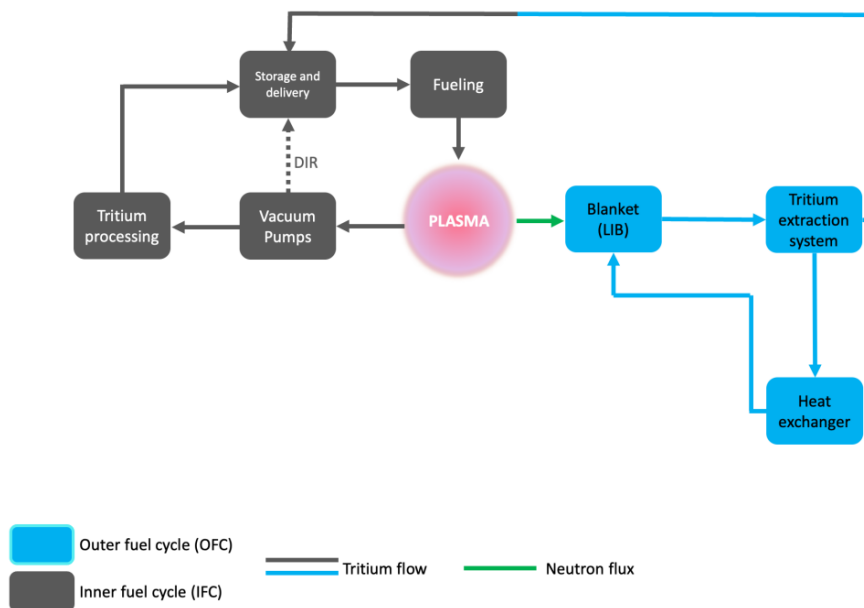


Figure 1: Simplified fuel cycle layout of a compact tokamak fusion power plant [1].

The general governing equation for the time-dependent tritium inventory in a generic component i is given by:

$$\frac{dI_i}{dt} = \sum_{j \neq i} \left(f_{j \rightarrow i} \frac{I_j}{\tau_j} \right) - (1 + \varepsilon_i) \left(\frac{I_i}{\tau_i} \right) - \lambda I_i + \dot{T}_i \quad (1)$$

In this equation, I_i denotes the tritium inventory [kg] in component i , τ_i is the corresponding residence time [s], and λ is the radioactive decay constant for tritium [s^{-1}]. The parameter ε_i represents the fraction of tritium lost from component i due to non-radioactive phenomena, such as leakages. The term $\dot{T}_i$ represents an external tritium source [kg s^{-1}] entering component i . The factor $f_{j \rightarrow i}$ defines the fraction of the outflow from component j that is directed to component i . The specific residence times for each fuel cycle component are provided. In Table 2 all the design parameters and residence times of each fuel cycle component are summarised.

In fact, the first term of Equation (1) represents the sum of the inflows to the component, while the second term corresponds to the outflows, including leakages. The third term accounts for the loss of tritium inventory in the component due to tritium decay, and the last term acts as either a sink or a source, depending on its sign (positive or negative, respectively).

Table 1: Residence times and design parameters of the fuel cycle components.

Data	Symbol	Value	Units
Breeding blanket	τ_{BB}	3600	s
Tritium extraction system	τ_{TES}	24	h
Heat exchanger	τ_{HX}	1	h
Tritium processing plant	τ_p	6	h
TES efficiency	η_{TES}	0.9	—
Non-radioactive tritium losses	ϵ	1×10^{-4}	—
Tritium burn efficiency	TBE	0.01	—
Direct internal recycling fraction	f_{DIR}	0.3	—

3 Calculations

This section outlines the calculations for each component, presented in the order in which they were performed during the exercise.

3.1 Primary vacuum pumping system

The analysis begins by estimating the number of cryogenic pumps required to exhaust unburnt tritium, deuterium, and helium ash from the divertor region. The divertor pressure is maintained at $p_{div} = 1 \text{ Pa}$ and the primary pumping system employs cryopumps with a real pumping speed for deuterium of $S_D^{real} = 50 \text{ m}^3 \text{ s}^{-1}$.

The pumping system must handle a throughput equal to the particle exhaust entering from the divertor, expressed as

$$Q = n_{pumps} \cdot p_{div} \cdot S \quad (2)$$

The incoming particle fluxes and associated throughputs are defined as:

$$\begin{cases} \dot{N}_{He,div} = \frac{P_{fus}}{E_{fus}} \\ Q_{He} = \dot{N}_{He,div} \cdot \frac{k_B T_0}{p_{div}} \end{cases} \quad \begin{cases} \dot{N}_{T,div} = \dot{N}_{D,div} = \frac{\dot{N}_{He,div}}{(TBE - 1)} \\ Q_T = Q_D = \dot{N}_{T,div} \cdot \frac{k_B T_0}{p_{div}} \end{cases} \quad (3)$$

$$Q_{tot} = Q_{He} + Q_T + Q_D \quad (4)$$

Here $k_B = 1.380 \times 10^{-23} \text{ J K}^{-1}$ is the Boltzmann constant and $T_0 = 273.15 \text{ K}$. Since the reference pumping speed is given for deuterium, the pumping speed for each species i is corrected using:

$$S_i = S_D^{real} \cdot \sqrt{\frac{MM_D}{MM_i}} \quad (5)$$

The divertor exhaust composition is assumed to be 0.55% helium, with the remaining 99.45% equally divided between deuterium and tritium. The assumption for the helium fraction at the divertor is taken considering a fraction of alpha particles in the core $f_{\alpha,core}$ equal to 0.055% and a helium enrichment coefficient η_{He} equal to 0.1 [2]. Denoting by x_i the fraction of each species, the effective pumping speed becomes

$$S_{eff} = x_T \cdot S_T + x_D \cdot S_D^{real} + x_{He} \cdot S_{He} \quad (6)$$

The number of cryopumps required is then computed from

$$n_{pumps} = \frac{Q_{tot}}{S_{eff} \cdot p_{div}} \quad (7)$$

and rounding conservatively to the next higher integer gives $n_{pumps} = 3$.

After determining the number of operating cryopumps, the residence time of the system (τ_{CP}), which is not known a priori, must be computed. This is evaluated based on the time required for the sorbent material to reach saturation and therefore trigger a regeneration cycle. The first step is to calculate the number of moles that a single pump can adsorb before regeneration, using the sorbent gas capacity ($C_{sorbent}$), its surface area density ($a_{sorbent}$), and the cryopanel area (A_{panel}), as:

$$\text{Mol}_{H,sat} = C_{sorbent} \cdot a_{sorbent} \cdot A_{panel} = 45 \text{ mol} \quad (8)$$

Knowing this and the molar flow rate of deuterium and tritium coming from the divertor, which can be evaluated as:

$$\dot{\text{Mol}}_H = \frac{\dot{N}_{D,div} + \dot{N}_{T,div}}{N_{av}} \quad (9)$$

where N_{av} is the Avogadro's number, the residence time can be calculated then as:

$$\tau_{CP} = n_{pumps} \cdot \frac{\text{Mol}_{H,sat}}{\dot{\text{Mol}}_H} \approx 2315.3 \text{ s} \quad (10)$$

The term for the number of pumps follows from the requirement that the total molar flow rate is distributed among all pumps, which are assumed to operate in parallel.

In practice, the number of pumps is usually at least twice the calculated minimum to ensure continuous operation. Figure 2 illustrates two consecutive working cycles, providing a simplified example of how continuous operation can be maintained while allowing pumps to regenerate.

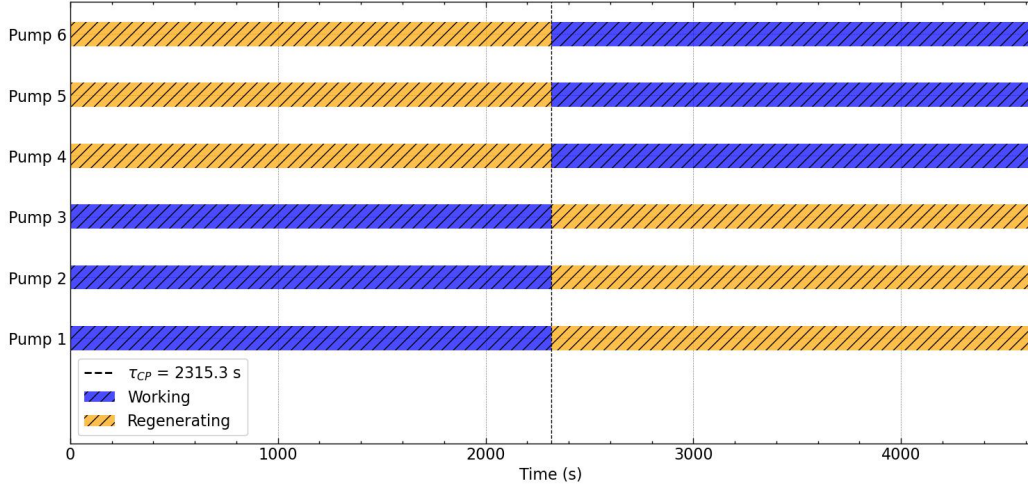


Figure 2: Two consecutive cycles of six cryogenic pumps. The diagram shows the staggered timing of the 'Working' (blue) and 'Regenerating' (orange) phases, assuming both phases are equal to the residence time.

A safety constraint is imposed on the cryopump design and needs to be checked. In a worst-case scenario involving a breach and air ingress into the cryopump volume, the resulting hydrogen (including all hydrogen isotopes) concentration must not exceed the Lower Flammability Limit (LFL). With a specified cryopump volume $V_{CP} = 10 \text{ m}^3$, and assuming that the air mixture is at $T = 300 \text{ K}$ and $p = 1 \text{ bar}$, the maximum allowable hydrogen inventory in the pump is determined by limiting the volume concentration to 3%.

To evaluate whether this limit is respected, we consider that the maximum number of moles of hydrogen trapped inside a single pump before regeneration is 45 mol. Using the ideal gas law, this molar quantity is then calculated to yield the maximum hydrogen volume inside the pump, which is given by:

$$V_{max} = \frac{\text{Mol}_{H,sat} \cdot R \cdot T}{p} = 1.1224 \text{ m}^3 \quad (11)$$

The volume obtained is then divided by the total volume of a single pump to calculate the concentration, which is $c = 11.22\%$. Since this value is well above the safety limit, the constraint is not respected. This implies that in a real application, regeneration must be performed before the residence time expires to ensure a lower accumulated hydrogen inventory in the pump.

Known the residence time, the equation for the cryopumps can be written:

$$\frac{dI_{CP}}{dt} = -(1 + \varepsilon) \cdot \left(\frac{I_{CP}}{\tau_{CP}} \right) - \lambda \cdot I_{CP} + \dot{T}_{T,div} \quad (12)$$

where

$$\dot{T}_{T,div} = \frac{\dot{N}_{T,div} \cdot \text{MM}_T}{N_{av}} \quad (13)$$

The source term $\dot{T}_{T,div}$ represents the mass flow rate of tritium entering the cryopump from the divertor. In the accompanying equation, the MM_T (molar mass of tritium) and N_{av} serve

as conversion factors to transform the input particle flow rate $\dot{N}_{T,div}$ (in particles/s) into the required mass flow rate $\dot{T}_{T,div}$ (in kg s⁻¹). From now on, the ratio MM_T/N_{av} will be called *conv*.

3.2 Fueling system and plasma exhaust

The plasma itself and the fueling system are not explicitly modelled, but they establish the boundary conditions for the tritium flow. This includes an injection flow from the fueling system (feeding the SDS) and an exhaust flow of unburned fuel and other species onto the primary vacuum pumping system. The required fueling rate ($\dot{N}_{inj}$) is consequently defined using the Tritium Burn Efficiency (TBE), which is the fractional measure of tritium undergoing fusion, as:

$$\dot{N}_{inj} = \frac{\dot{N}_{T,burn}}{TBE} = \frac{\dot{N}_{He,div}}{TBE} \quad (14)$$

The plant operates on a pulsed schedule, with a pulse duration $t_{pulse} = 15$ min followed by a dwell time $t_{dwell} = 2$ min.

3.3 Tritium processing system

For the tritium processing system, the equation is the following:

$$\frac{dI_P}{dt} = (1 - f_{DIR}) \cdot \left(\frac{I_{CP}}{\tau_{CP}} \right) - (1 + \varepsilon) \cdot \left(\frac{I_P}{\tau_P} \right) - \lambda \cdot I_P \quad (15)$$

Here, it can be seen that a fraction of the flow comes from the cryopumps (first term), with a value of $1 - f_{DIR}$. This represents the complement of the f_{DIR} fraction, since part of the flow from the cryopumps is directed to the tritium processing system, while the remaining portion is routed to the storage and delivery system through Direct Internal Recycling (DIR).

3.4 Storage and delivery system

The same procedure as before can also be applied to the storage and delivery system, obtaining the following equation:

$$\frac{dI_{SDS}}{dt} = f_{DIR} \cdot \left(\frac{I_{CP}}{\tau_{CP}} \right) + 1 \cdot \left(\frac{I_P}{\tau_P} \right) + \eta_{TES} \cdot \frac{I_{TES}}{\tau_{TES}} - \lambda \cdot I_{SDS} - \dot{T}_{T,inj} \quad (16)$$

As mentioned, in this component, only the fraction f_{DIR} is coming from the cryopumps. For the flow coming from the tritium processing plant, instead, the fraction is unitary, since all the quantity that exit from the processing plant is sent to the storage and delivery system. Another fraction is coming from the tritium extraction system, which is equal to the TES efficiency (η_{TES}). Moreover, the loss term (ε) is neglected since it is assumed to have no losses in the SDS. $\dot{T}_{T,inj}$ is a sink term which account for the tritium injected into the plasma to sustain the fusion reaction and is evaluated as:

$$\dot{T}_{T,inj} = \dot{N}_{T,inj} \cdot conv \quad (17)$$

3.5 Breeding blanket

The breeding blanket is a critical component for tritium self-sufficiency. The rate of tritium production in the blanket is:

$$\dot{T}_{BB} = \dot{N}_{He,div} \cdot TBR_r \cdot conv \quad (18)$$

Here, TBR_r is the required Tritium Breeding Ratio, which is one of the key unknowns to be determined through dynamic simulation. All the tritium produced in the blanket is carried to the tritium extraction system (TES), giving as final equation:

$$\frac{dI_{BB}}{dt} = 1 \cdot \left(\frac{I_{HX}}{\tau_{HX}} \right) - (1 + \varepsilon) \cdot \left(\frac{I_{BB}}{\tau_{BB}} \right) - \lambda \cdot I_{BB} + \dot{T}_{BB} \quad (19)$$

The TES processes the stream coming from the breeding blanket with a certain efficiency, meaning a fraction η_{TES} of the incoming tritium is successfully extracted and sent to the SDS, while the remaining fraction is directed to the heat exchanger and then reinjected into the BB. This can be seen expressed in the first term of the equation above, with the inflow coming from the HX.

3.6 Tritium extraction system

The final equation for the tritium extraction system is:

$$\frac{dI_{TES}}{dt} = 1 \cdot \left(\frac{I_{BB}}{\tau_{BB}} \right) - (1 + \varepsilon) \cdot \left(\frac{I_{TES}}{\tau_{TES}} \right) - \lambda \cdot I_{TES} \quad (20)$$

Since all the tritium entering the TES is coming from the breeding blanket, the first term has a unitary coefficient in front. The remaining terms are related to outflows and leakages.

3.7 Heat exchanger

The heat exchanger is a simple component with only one inflow and one outflow and the equation is the following:

$$\frac{dI_{HX}}{dt} = (1 - \eta_{TES}) \cdot \left(\frac{I_{TES}}{\tau_{TES}} \right) - (1 + \varepsilon) \cdot \left(\frac{I_{HX}}{\tau_{HX}} \right) - \lambda \cdot I_{HX} \quad (21)$$

In the first term, the $(1 - \eta_{TES})$ coefficient account for the fraction of tritium coming from the tritium extraction system, which is the one that do not go to the breeding blanket.

3.8 Dynamic simulation for inventory and TBR determination

The core of the analysis involves simulating the transient and steady-state behaviour of the entire fuel-cycle system. An initial value is assumed for the startup inventory $I_{startup}$, which is applied directly as the initial condition of the SDS, and this value is adjusted until the minimum of the SDS inventory evolution becomes as close as possible to zero. An initial value is also assumed for the required Tritium Breeding Ratio TBR_r , which is then iteratively modified in order to achieve the target doubling time t_d of 2 years.

In order to solve the Ordinary Differential Equations (ODEs) the *integrator* block on *Simulink* has been used [3]. As all the startup inventory is located in the Storage and delivery system, except it, all the other components of the cycle have the initial condition of their

integrator block set to 0. To evaluate the imposed inflows and outflows coming from the fusion reaction ($\dot{T}_{BB}$, $\dot{T}_{T,inj}$, $\dot{T}_{T,div}$), a plasma source has been directly simulated. To do this, a square function generator has been set up, providing a pulse with an amplitude of 500 MW and width of 15 min and then a dwell where power goes to zero of 2 min, as shown in Figure 3.

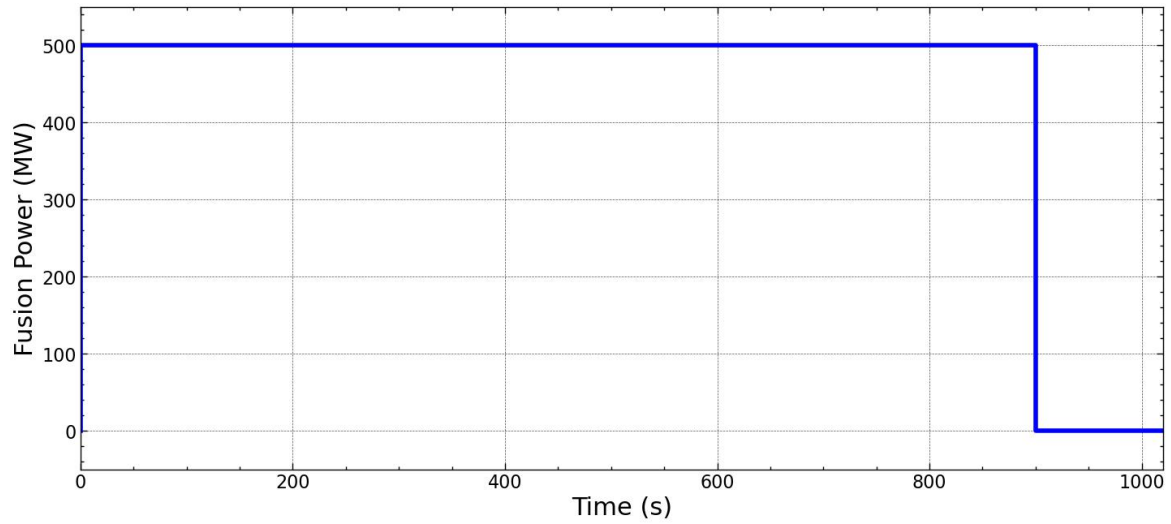


Figure 3: One period of fusion power square function.

4 Results

Starting with an initial guess of $TBR_r = 1.04$, the minimum startup inventory was determined by iterations in order to ensure that the SDS never depletes during the initial transient. For each startup guess, when the minimum TBR_r to reach the target inventory in 2 years was got, a verification on the minimum value of the inventory was made. If it was negative, a new iteration with a bigger startup started. Starting iterations with an initial guess of 1.3 kg, the optimal value was found to be $I_{startup} = 1.45$ kg.

It was known at that point that the design objective was to reach, during the ascent phase, an inventory target of 2.90 kg (i.e. the doubling of the initial 1.45 kg startup mass) within a doubling time $t_d = 2$ years.

In this perspective, Figure 4 displays the evolution of the SDS inventory over a 2-year period for different values of TBR_r . For TBR_r values between 1.04 and 1.08, tritium production is insufficient to meet the doubling target within the prescribed timeframe. The target of 2.90 kg after a time of 2 years is met with a $TBR_r = 1.09$. The minimum value of SDS inventory is reached at the inflection point, after almost 3 days, and it is equal to 34.98 g.

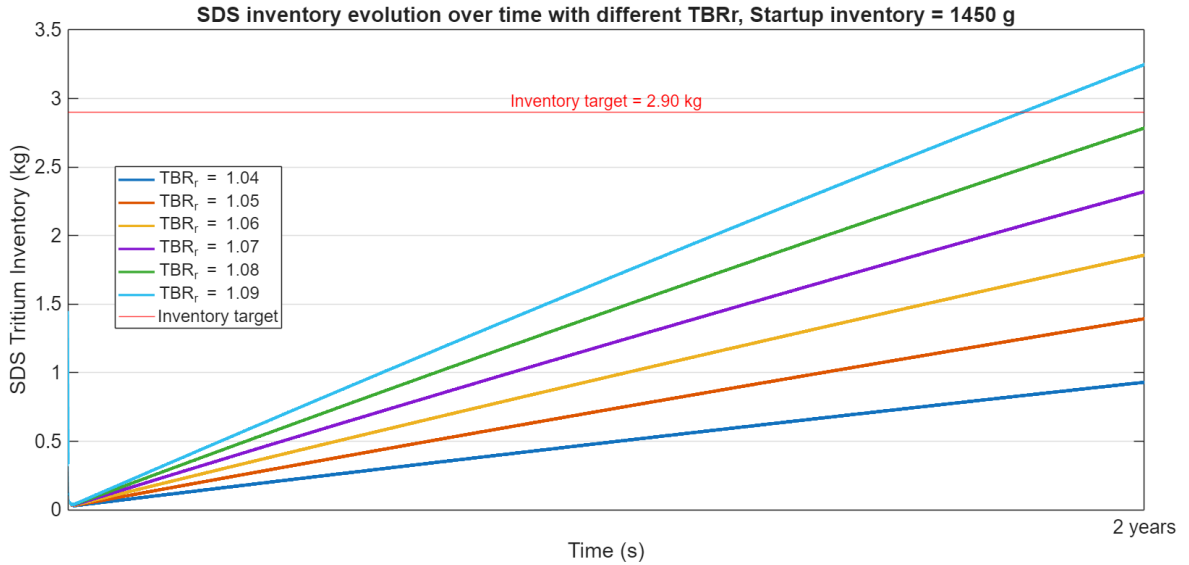


Figure 4: SDS inventory [kg] for different TBR_r over a 2 year period of time.

Then, the dynamic simulation of the tritium fuel cycle was performed again by integrating the mass balance equations (Eq. (1)) over a time horizon sufficient to observe both the startup transient and the final steady-state. This time was estimated to be $5 \cdot \tau_{TES} = 5$ days = 432000 s, as τ_{TES} is the bigger residence time. Transients of each component of the cycle are shown in Figures 5 and 6 and values at the steady-state are reported in Table 2.

Several considerations can be drawn from these results. The tritium processing system accounts for the largest share of the plant inventory, stabilizing at 1166.6 g; this is consistent with the high residence time ($\tau_p = 6$ h) required for isotopic separation, which leads to significant tritium holdup. Simultaneously, the vacuum pump inventory exhibits an oscillatory behavior ranging between 174 g and 184 g, reflecting the cryopump regeneration cycles described in Section 3.1 where tritium is periodically released from the sorbent panels. Regarding the tritium extraction system, the TES inventory stabilizes at approximately 80 g, a value governed by the extraction efficiency ($\eta_{TES} = 0.9$) and the characteristic residence time of 24 h. In contrast, the blanket maintains a very low inventory of 3.4 g due to the continuous and efficient removal of

the bred tritium from the FLiBe carrier. Finally, during the first hours of operation, the storage and delivery system (SDS) inventory undergoes a sharp depletion, representing the "charging" phase of the cycle. During this stage, tritium is transferred from storage to fill the various subsystems until equilibrium is reached and the steady-state inflection point is achieved.

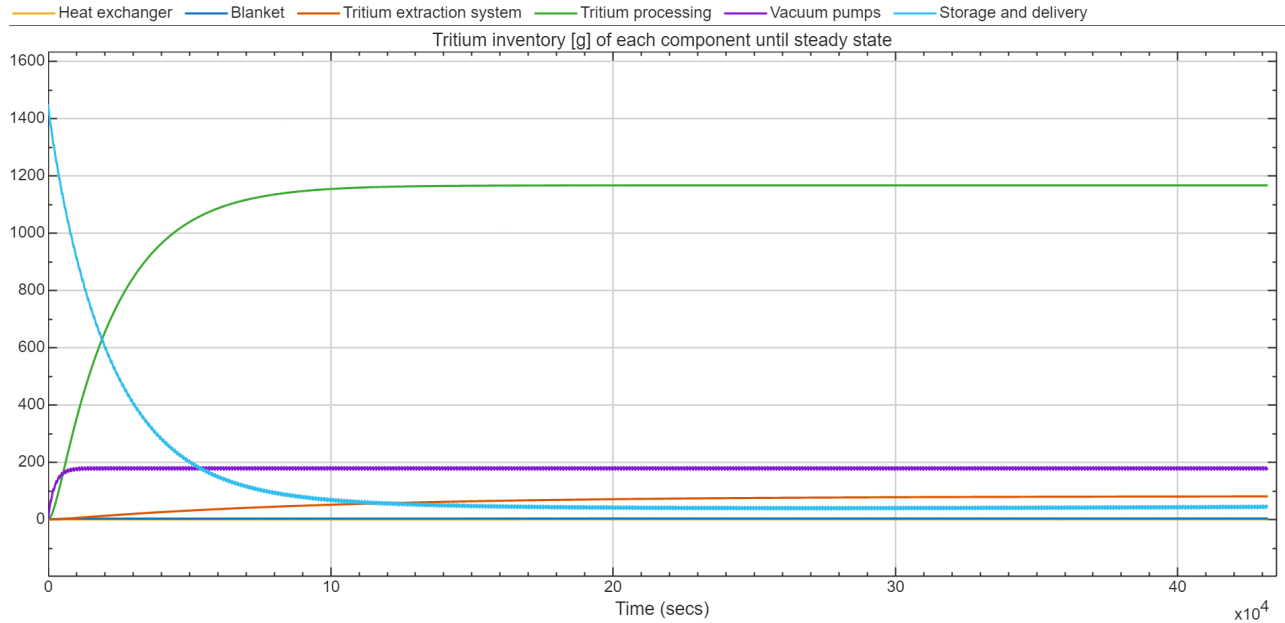


Figure 5: Tritium inventory [gram] at steady-state - one graph.

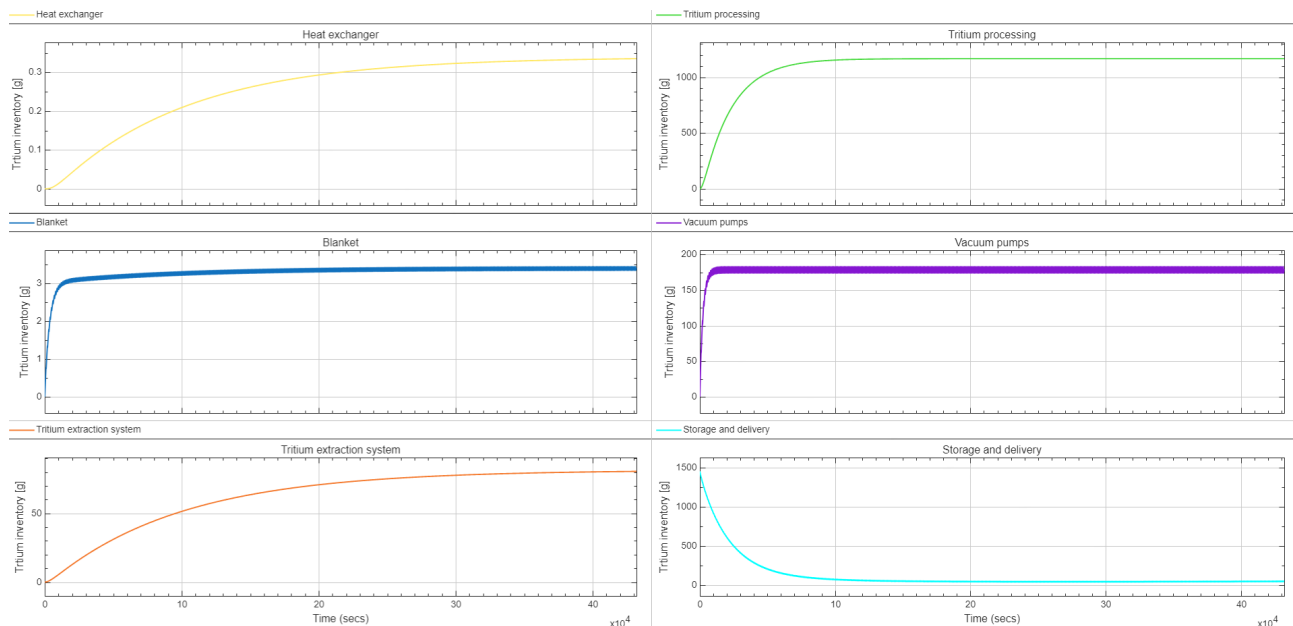


Figure 6: Tritium inventory [gram] at steady-state - separated graphs.

Table 2: Inventory values at steady-state for each component of the fuel cycle.

Cycle component	Inventory value at steady-state [g]
Vacuum pumps	179
Tritium processing	1166.6
Storage and delivery	44
Blanket	3.4
Tritium extraction system	80.5
Heat exchanger	0.34

5 Conclusions

This report provided a preliminary design and dynamic characterization of a closed D-T fuel cycle for a 500 MW_{th} compact tokamak. Through the implementation of a lumped parameter model on Simulink, the study successfully evaluated the tritium throughput and inventory evolution across the plant's primary subsystems.

The sizing of the primary vacuum pumping system determined that a minimum of three operational cryopumps is required to handle the divertor exhaust. However, the safety analysis regarding the Lower Flammability Limit (LFL) underscored a critical design constraint: to maintain the hydrogen isotope concentration below the 3% threshold, regeneration cycles should be performed more frequently than the theoretical saturation time of almost 2315 s.

Regarding fuel self-sufficiency, the dynamic simulations identified a minimum startup tritium inventory ($I_{startup}$) of 1.45 kg to ensure continuous operation and prevent depletion during the initial transient phase. To achieve the strategic objective of doubling this inventory within a two-year timeframe ($t_d = 2$ years), the required Tritium Breeding Ratio (TBR_r) was calculated to be 1.09.

Finally, the steady-state analysis revealed that the tritium processing system accounts for the largest share of the plant's total inventory (1166.6 g), primarily due to the significant residence time required for isotopic separation. These results highlight the fundamental interdependence between component residence times, safety constraints, and breeding performance, providing a quantitative foundation for the optimization of fuel cycles in compact fusion power plants.

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